Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/A_DEB_SAAWKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3333, C2 = 3663, C3 = 2212)

text_priors <- '
Distrib (Fm,           TruncNormal,           0.1,      0.2,  0.01,      0.8);
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40,    2000);
Distrib (r_pAm_pM,     TruncNormal_cv,        0.8,      0.1,    0.1,      2);
Distrib (v,            TruncNormal_cv,    0.18505,      0.1,  0.001,    2);
Distrib (kap,          Uniform,               0,     1);
Distrib (kapA,         Uniform,               0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40,    300);
Distrib (E_coc,        TruncNormal_cv,          30,      0.2,     1,    300);
Distrib (rOM_ClxHorse, LogUniform,         0.00001,      0.3);
Distrib (K,            Uniform,            0.00001,        2);
Distrib (dv,           TruncNormal_cv,           2,      0.5,     0,     10);
Distrib (wE,           LogUniform,      0.000000001,     0.1);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcali DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

```
                     Fm.1.    pAm.1. r_pAm_pM.1.      v.1.     kap.1.    kapA.1.
```

```
Result_mode 0.010213900 191.96400  0.77092900 0.1950370 0.94416500 0.56889200
2.5%        0.010012900 173.56300  0.70510900 0.1603526 0.92020230 0.50718700
97.5%       0.012011100 220.34025  0.93721050 0.2288020 0.96692520 0.67729300
Min         0.010000000  94.58760  0.63217100 0.1356970 0.50515300 0.18063900
Max         0.346466000 238.63600  0.98258200 0.2644820 0.97706800 0.73462200
Result_mean 0.010559066 198.14456  0.81899541 0.1934209 0.94824109 0.60257476
Result_sd   0.002756721  12.16692  0.05708689 0.0173446 0.01287656 0.04208748
               Ehp.1.  E_coc.1. rOM_ClxHorse.1.       K.1.     dv.1.
Result_mode 112.01600 39.347000    0.0089761200 0.04798550 2.7666700
2.5%         69.22760 29.848440    0.0056706900 0.03217210 2.3258700
97.5%       141.20900 44.835300    0.0115823000 0.06780797 3.4907400
Min          58.16190 24.503700    0.0000161998 0.02383290 0.5220400
Max         169.81200 50.453500    0.0155816000 0.98193800 4.5834400
Result_mean 103.76818 37.013124    0.0084990159 0.04762896 2.8492805
Result_sd    19.17282  3.887599    0.0015220541 0.01151621 0.3067326
                   wE.1. Sigma_W.1.
Result_mode 1.408640e-09 0.13496000
2.5%        1.286050e-09 0.12136500
97.5%       4.908717e-06 0.15531900
Min         1.005580e-09 0.11035600
Max         1.012430e-05 5.72094000
Result_mean 5.737672e-07 0.13766316
Result_sd   1.304946e-06 0.05374956

Number of observations, n =  277
Number of parameters, k =  12
Log-likelihood, LL =  -41.76121
BIC =  151.0106


Potential scale reduction factors:

                Point est. Upper C.I.
dv.1.                 1.16       1.48
E_coc.1.              1.08       1.24
Ehp.1.                1.11       1.33
Fm.1.                 1.02       1.05
K.1.                  1.19       1.54
kap.1.                1.35       2.01
kapA.1.               1.52       2.37
pAm.1.                1.20       1.57
r_pAm_pM.1.           1.40       2.04
```

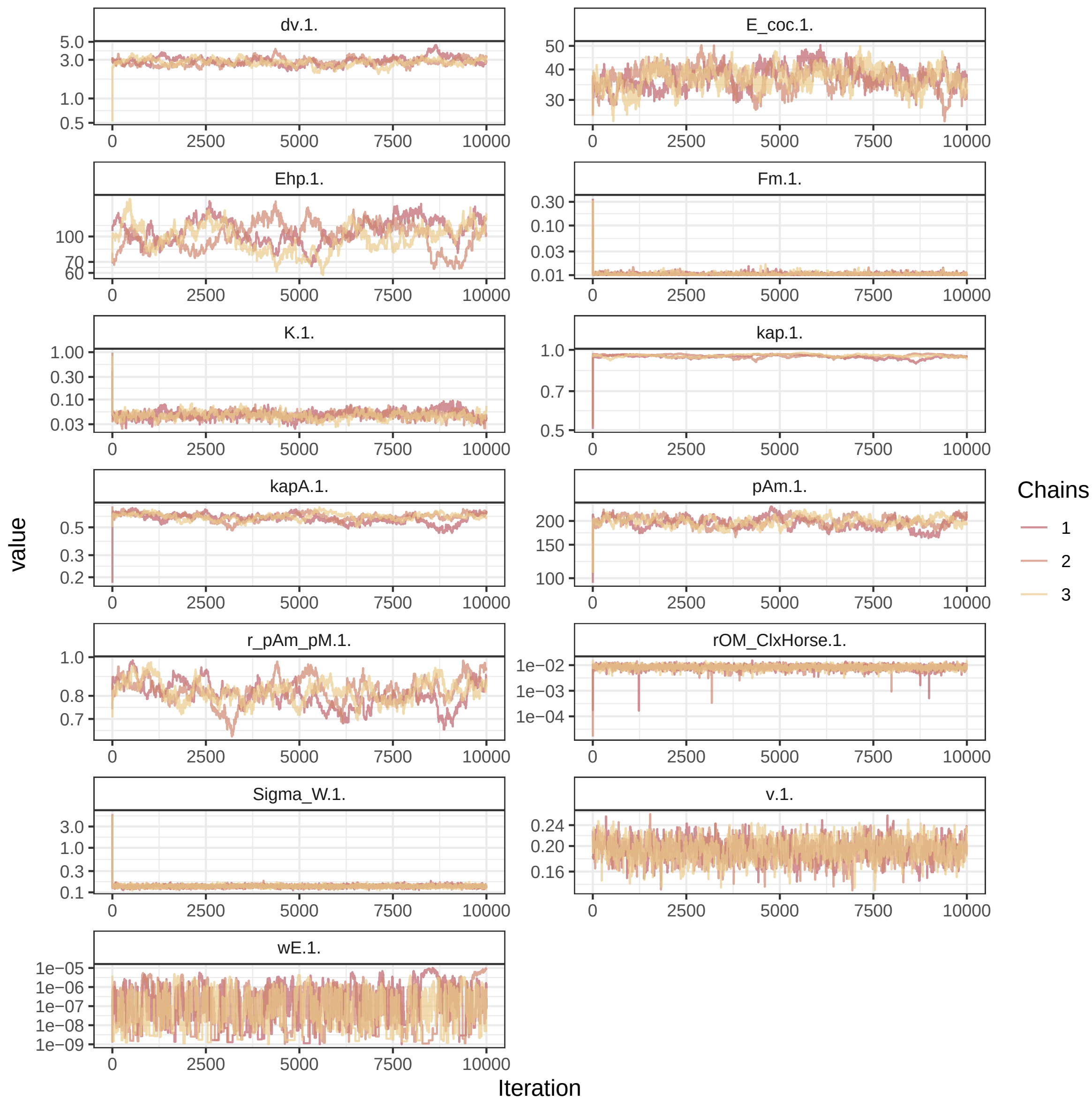


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

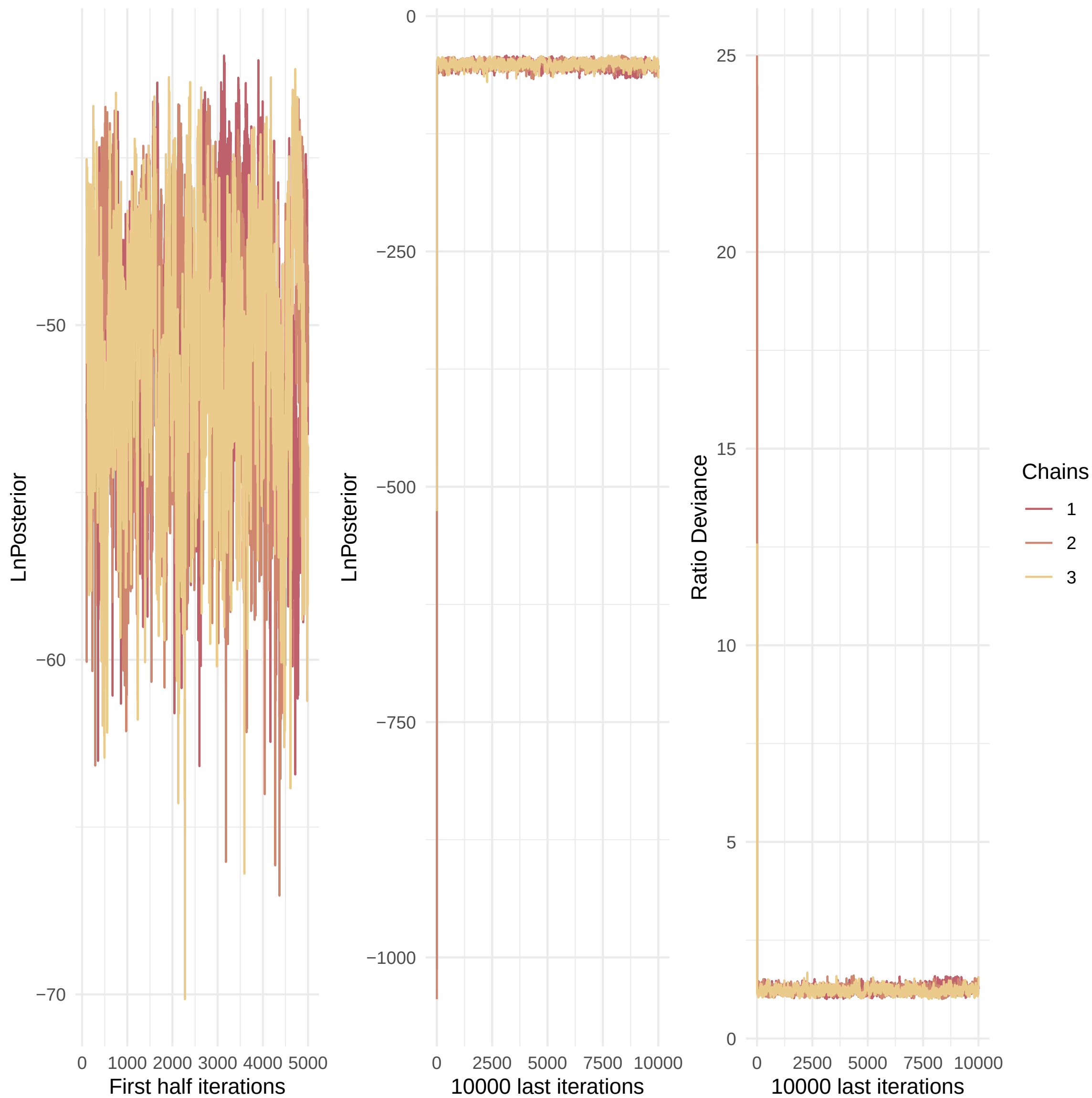


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

```
rOM_ClxHorse.1.       1.00       1.01
Sigma_W.1.            1.02       1.05
v.1.                  1.01       1.03
wE.1.                 1.13       1.34

Multivariate psrf

1.4
```

```r
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

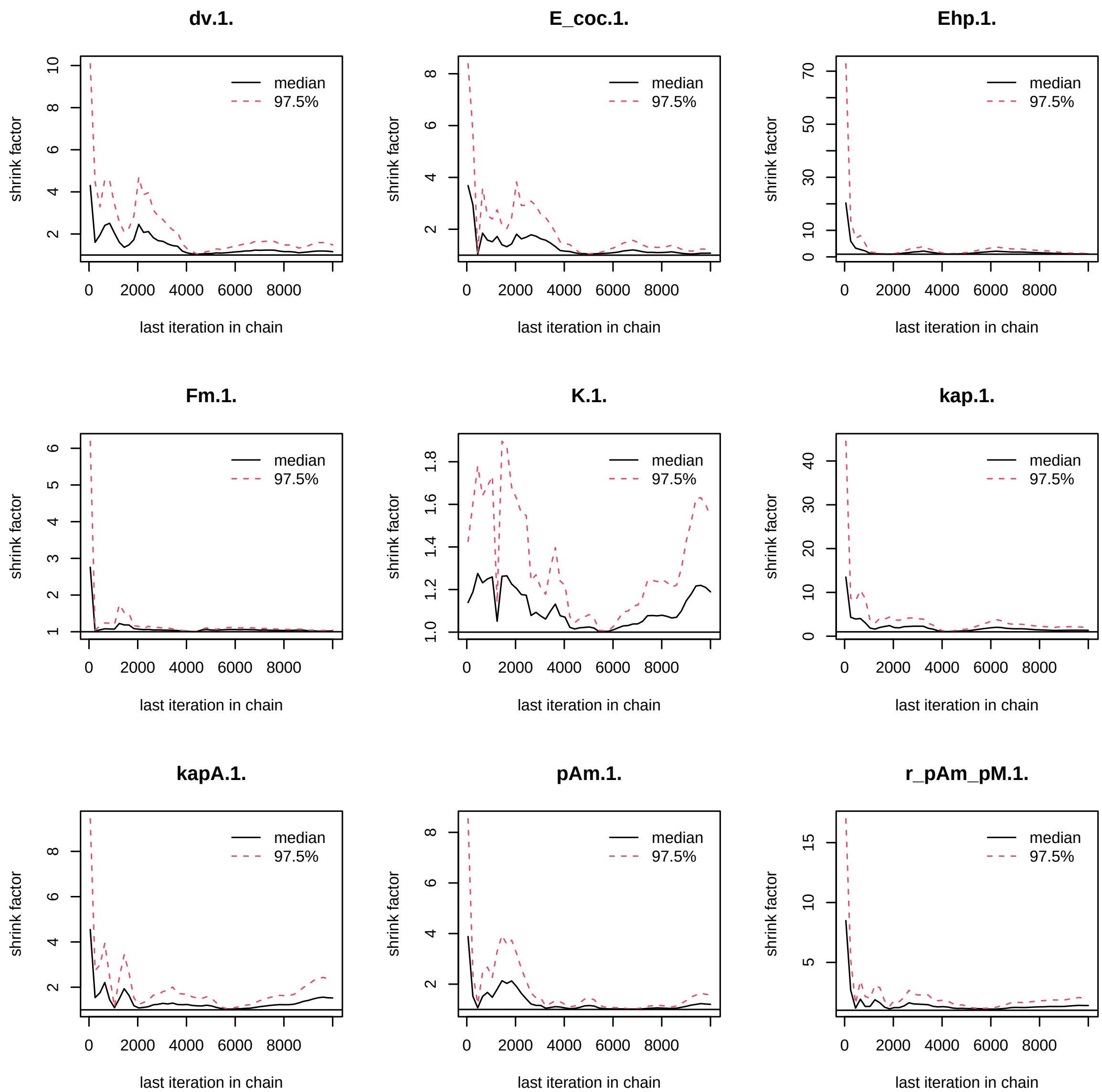


Figure 3: Chains convergence.

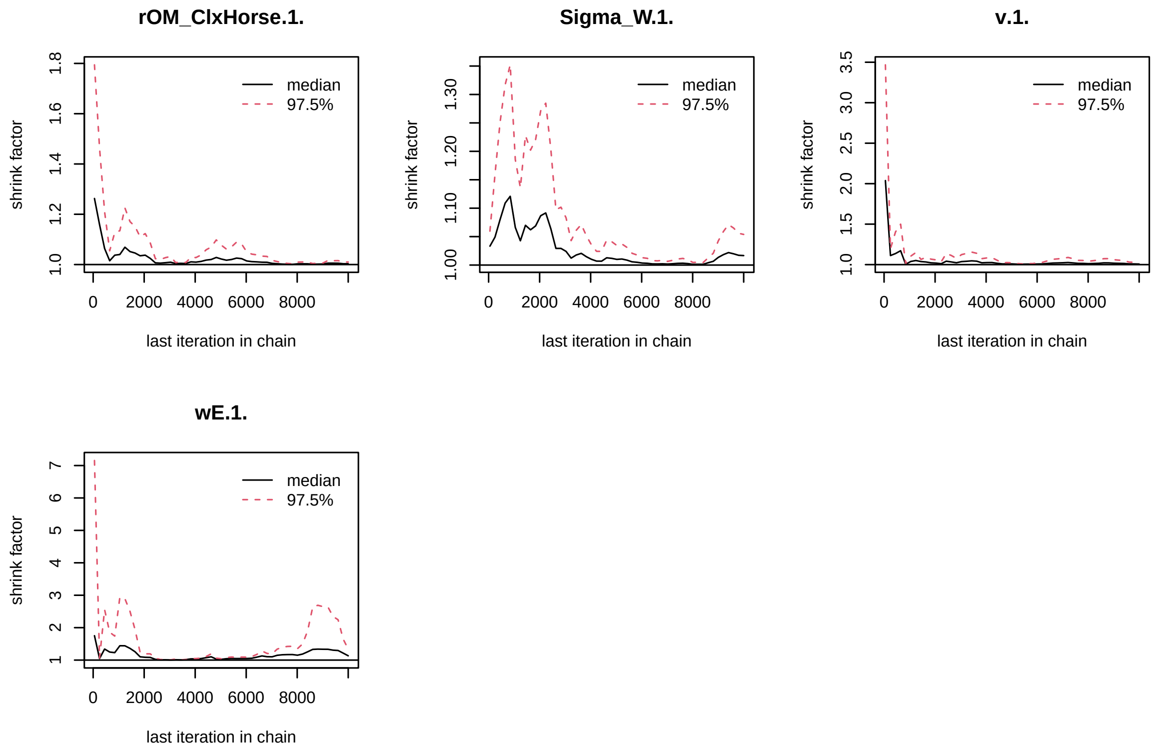


Figure 4: Chains convergence.

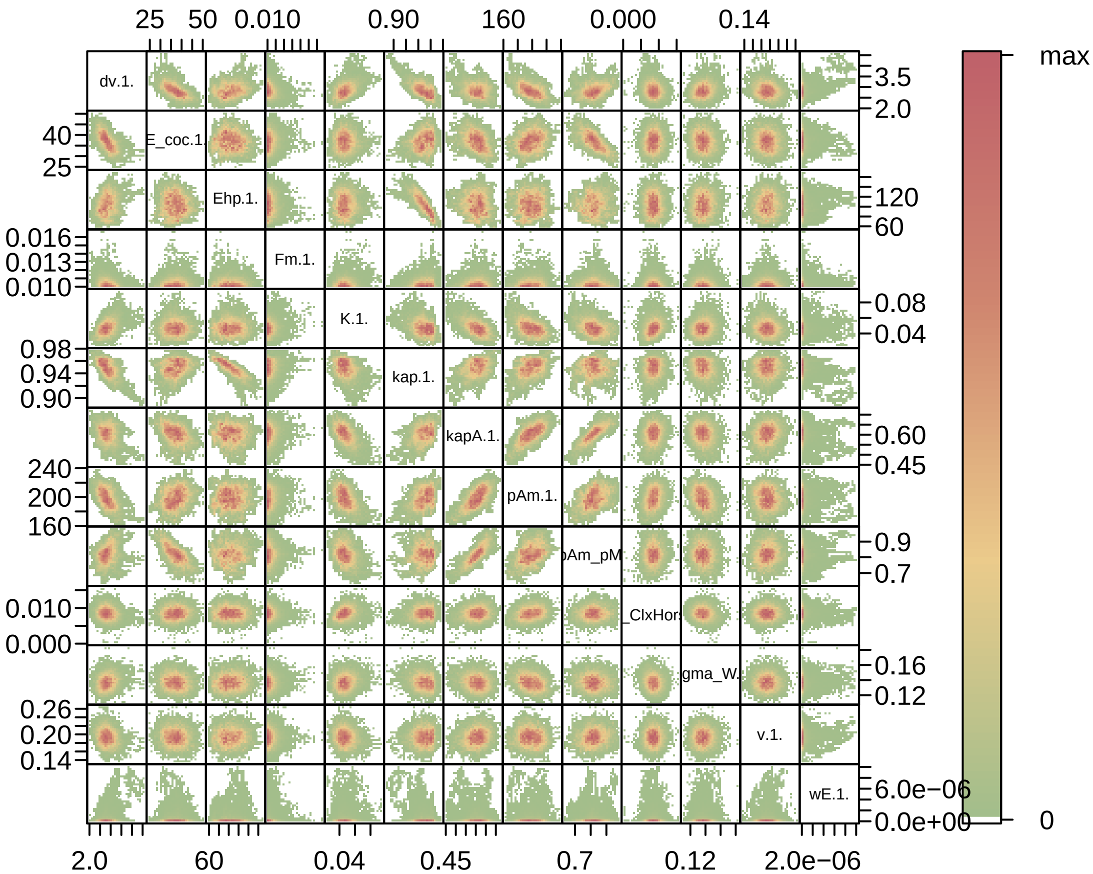


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

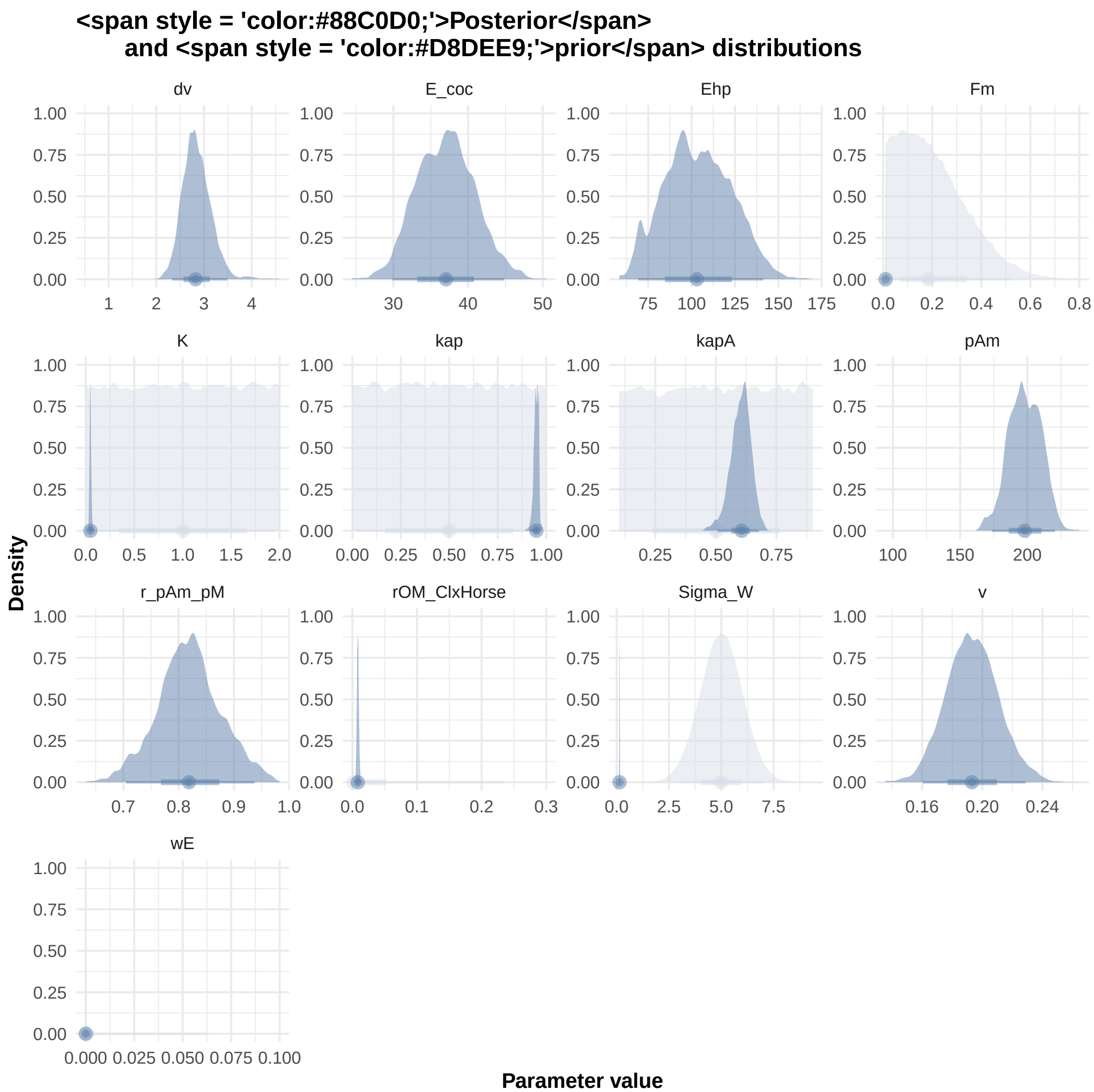


Figure 6: Distributions of the priors and the posteriors.

```
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,
    kapreal,
    freal,
    Qf_soil,
    Qf_horse,
    X"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in
```

```r
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
```

```
"many-to-many"` to silence this warning.
```

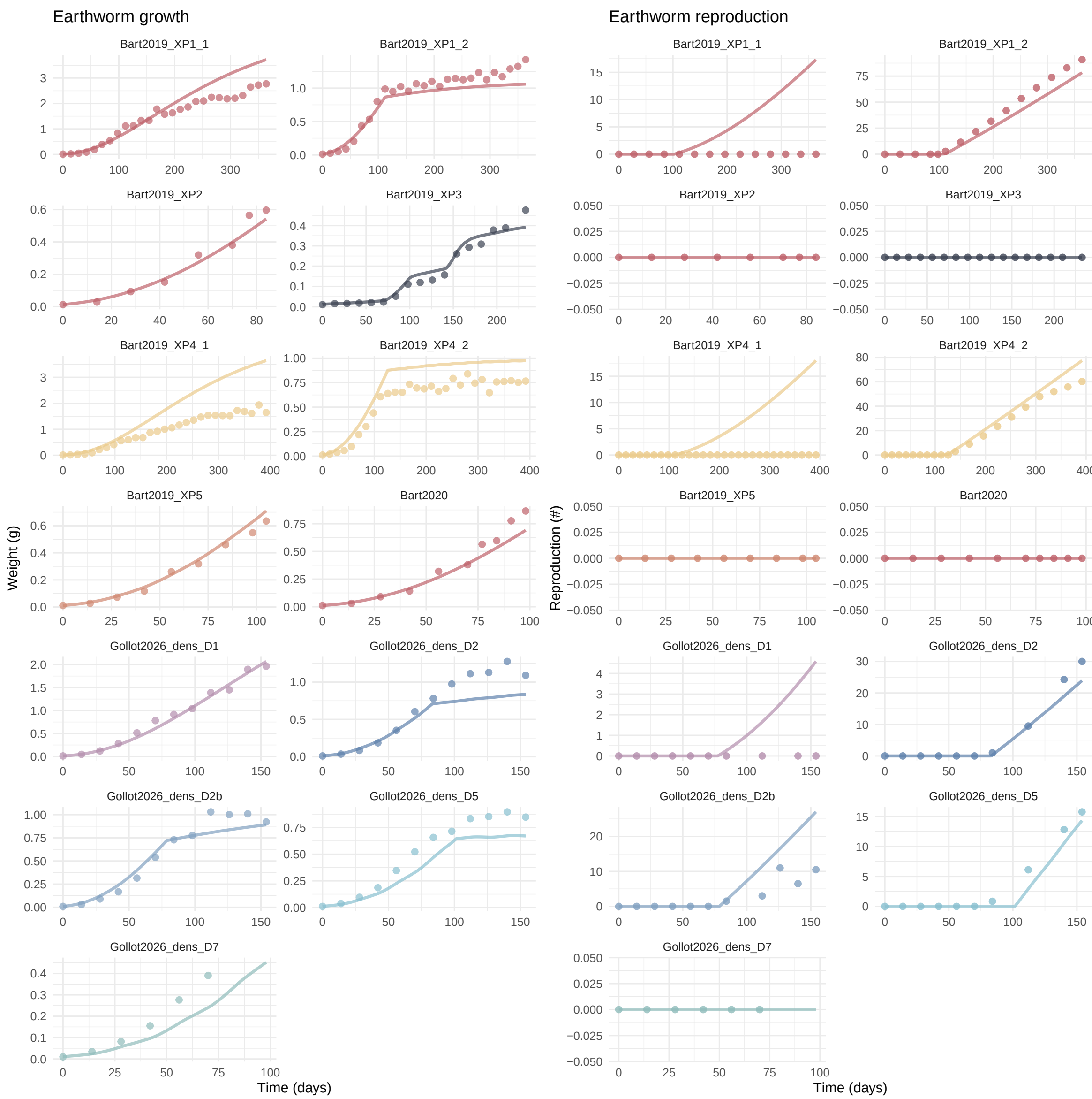
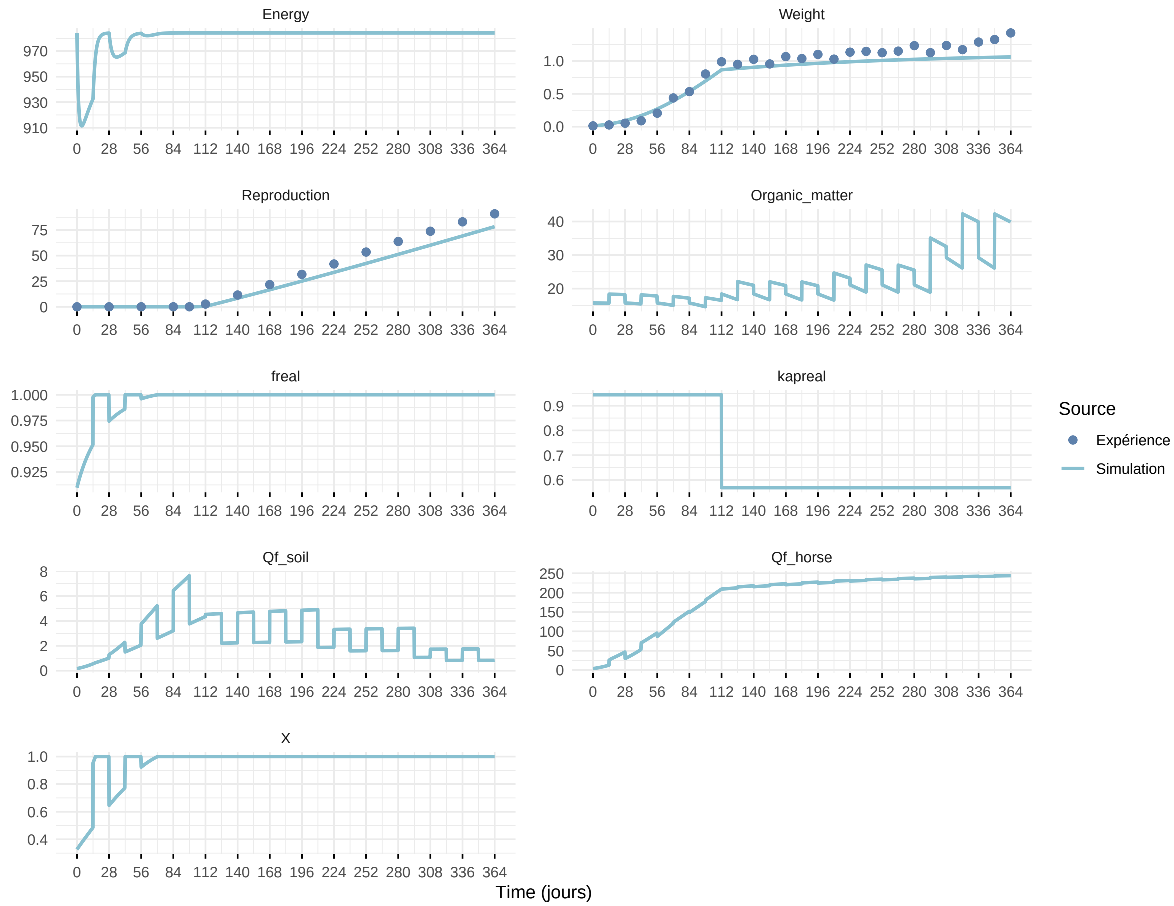
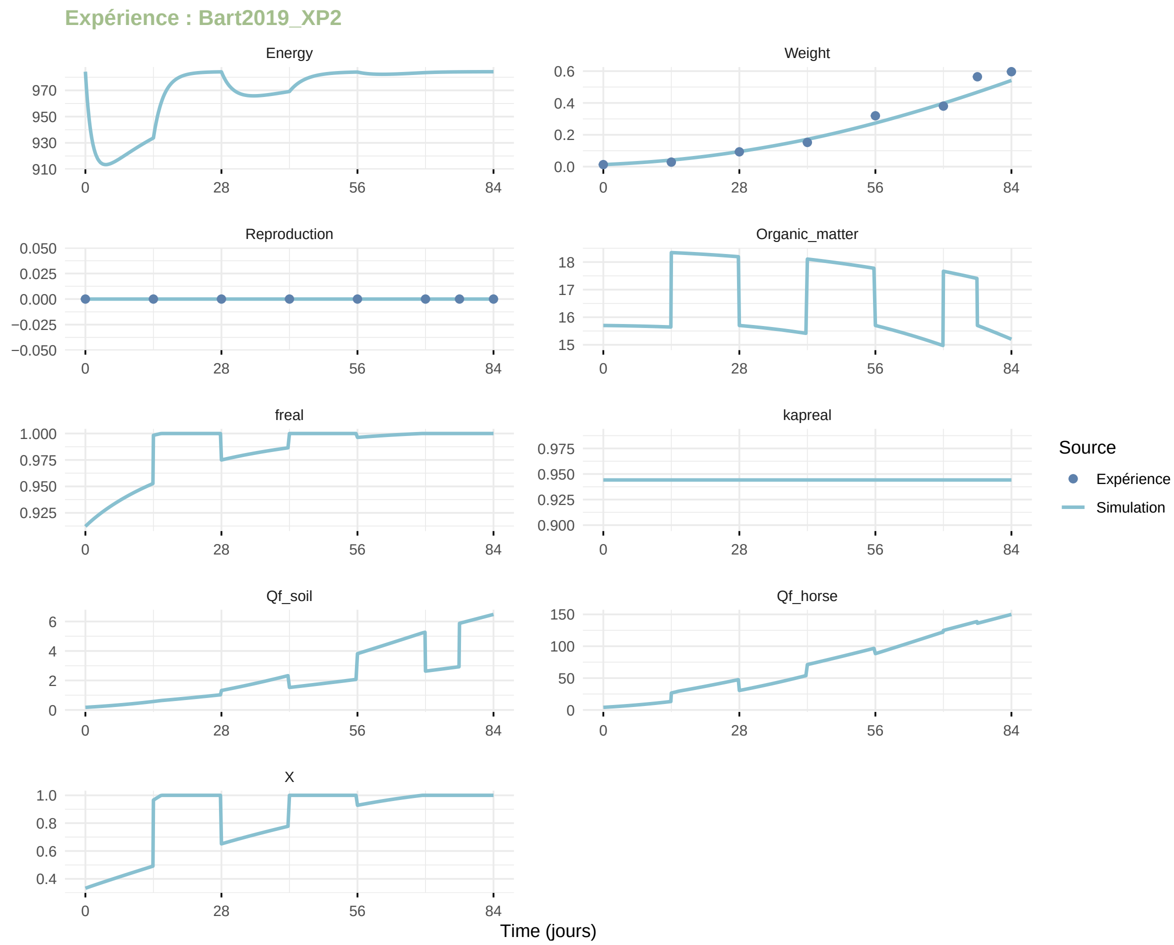
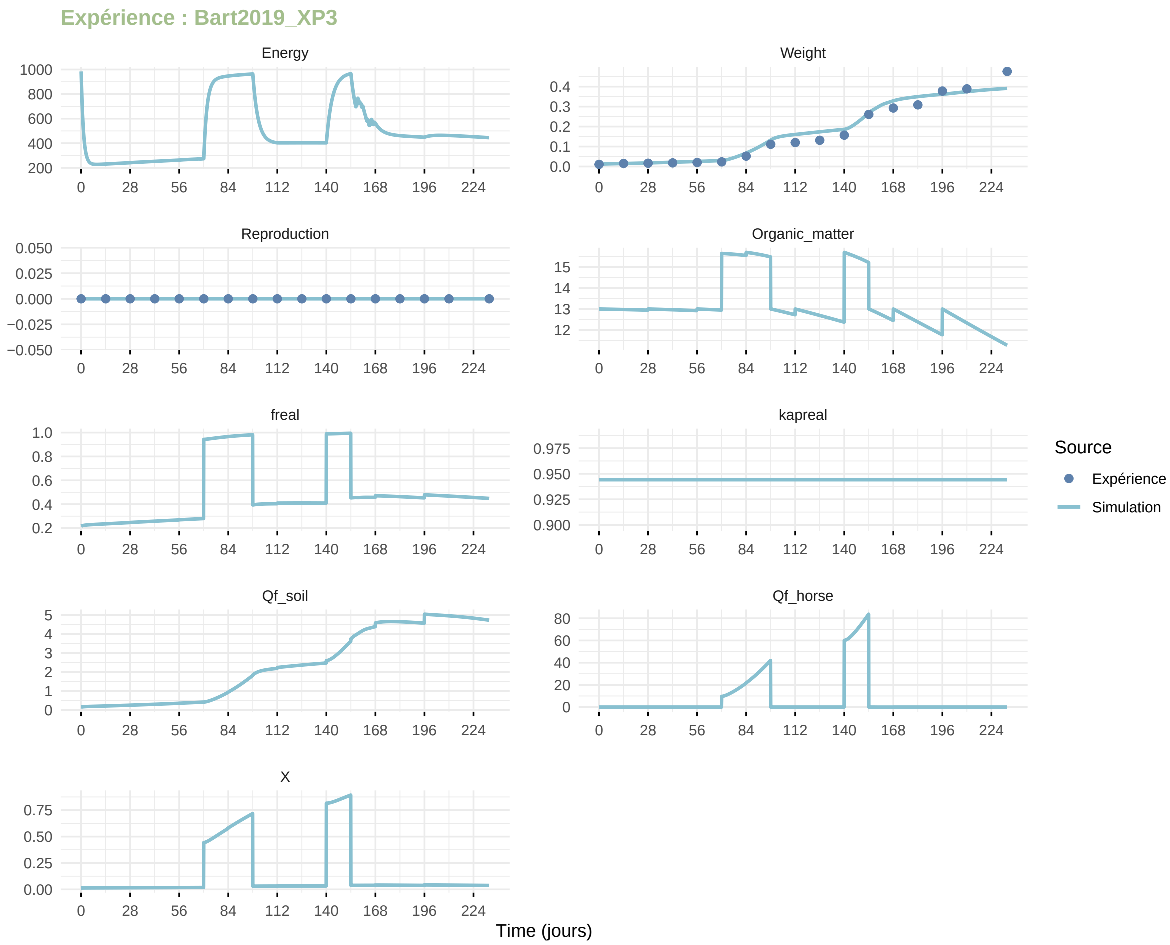


Figure 7: Simulations Weight and Reproduction.

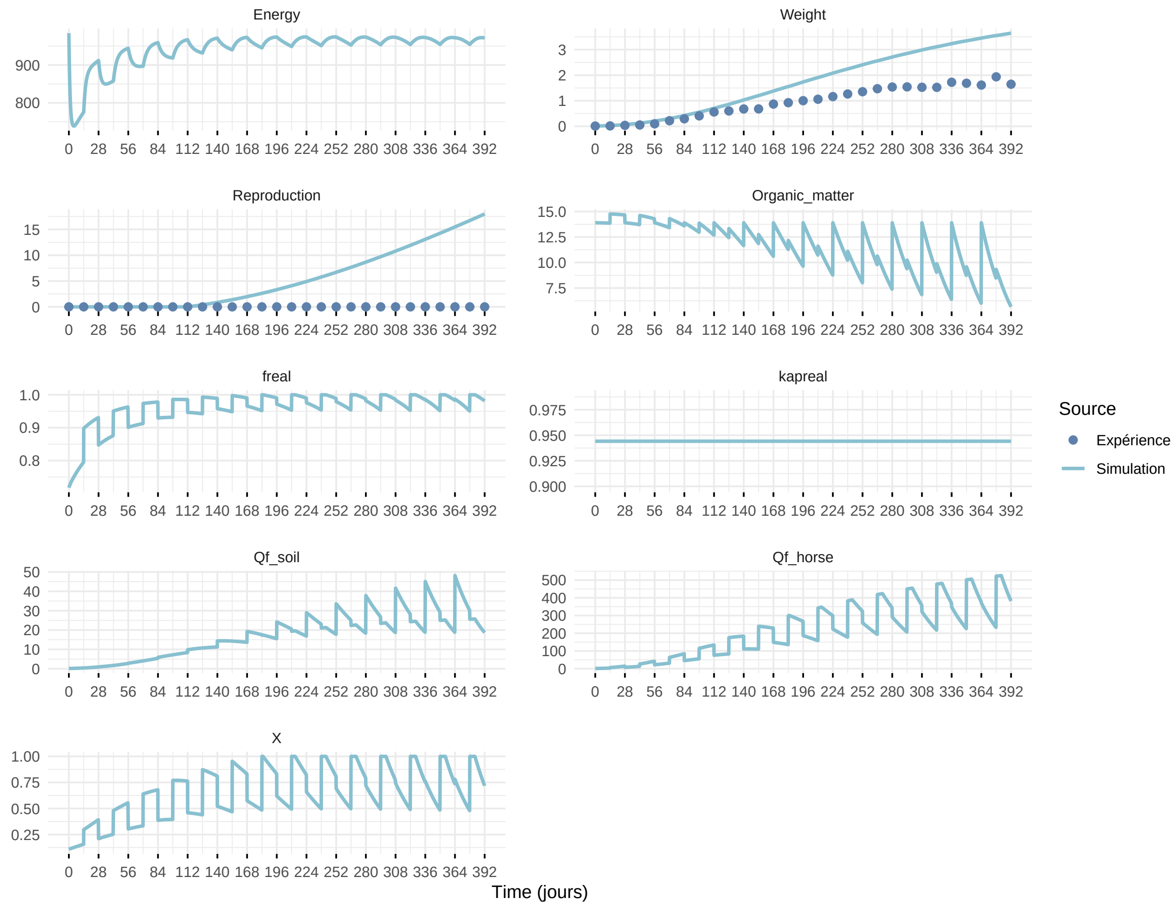
Expérience : Bart2019_XP1_2



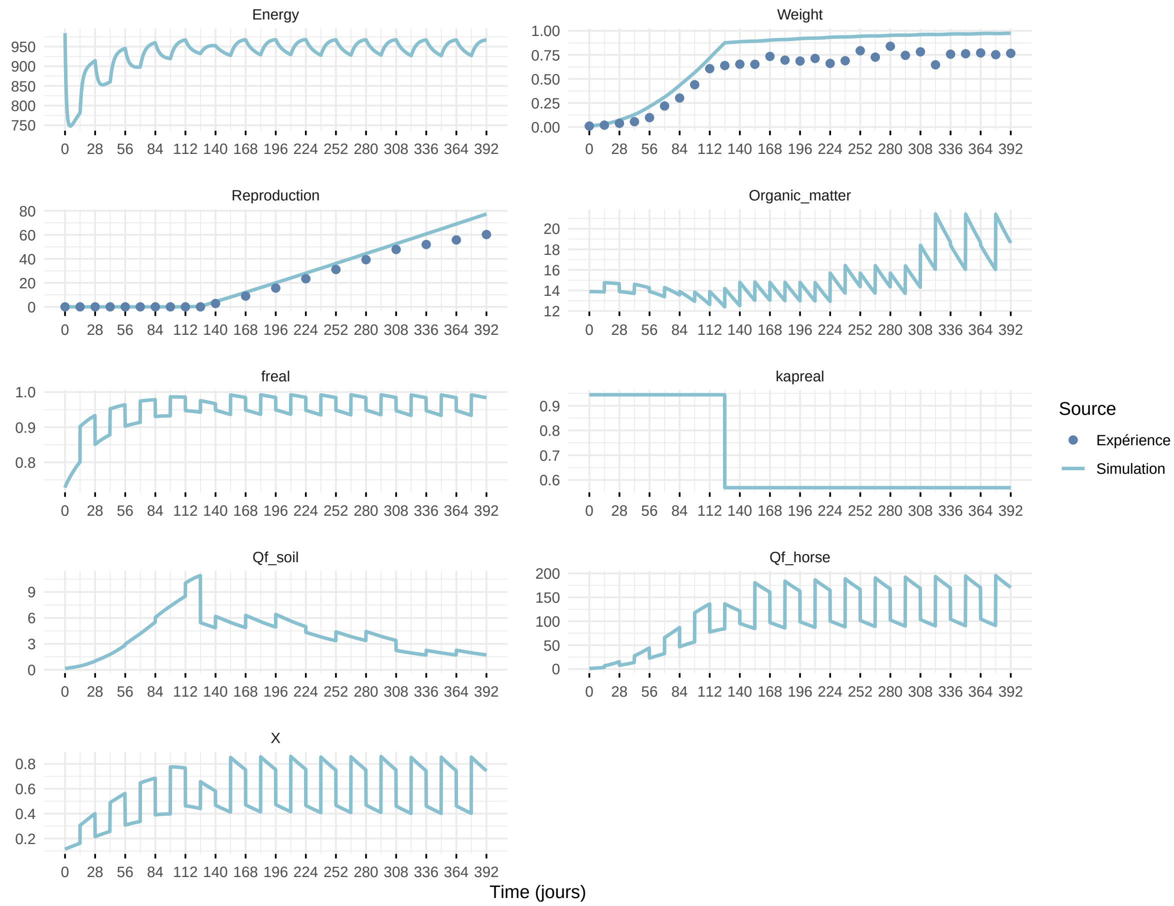

Expérience : Bart2019_XP2
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)

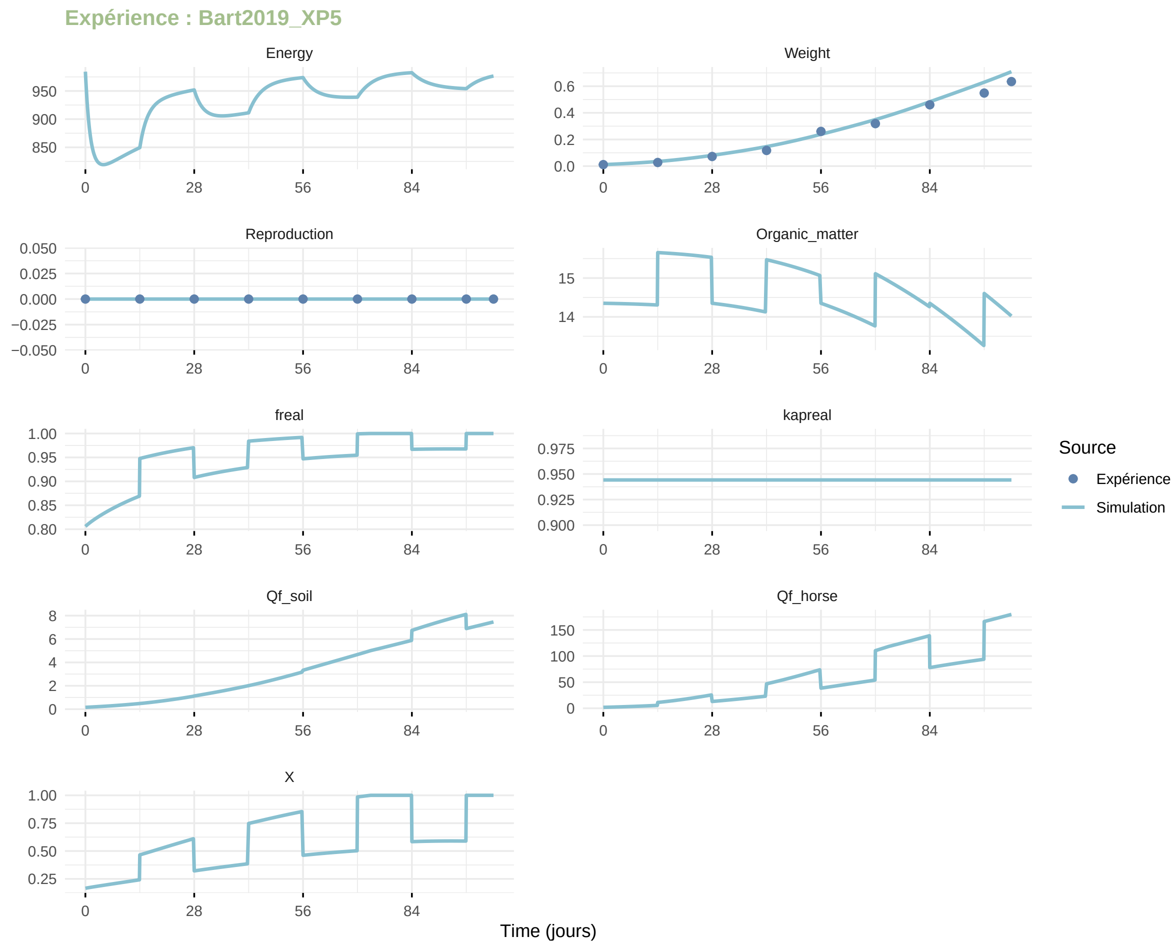

Expérience : Bart2019_XP3
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)

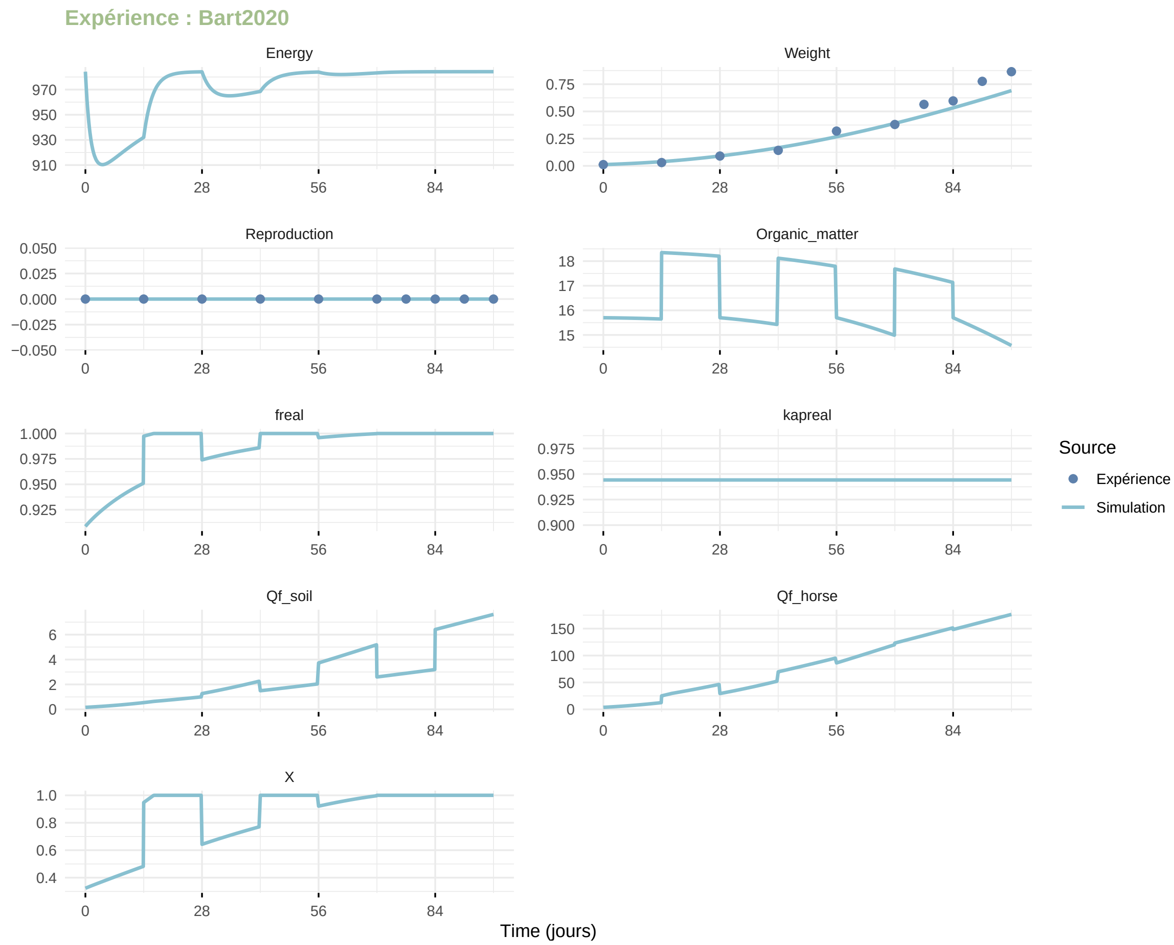
Expérience : Bart2019_XP4_1



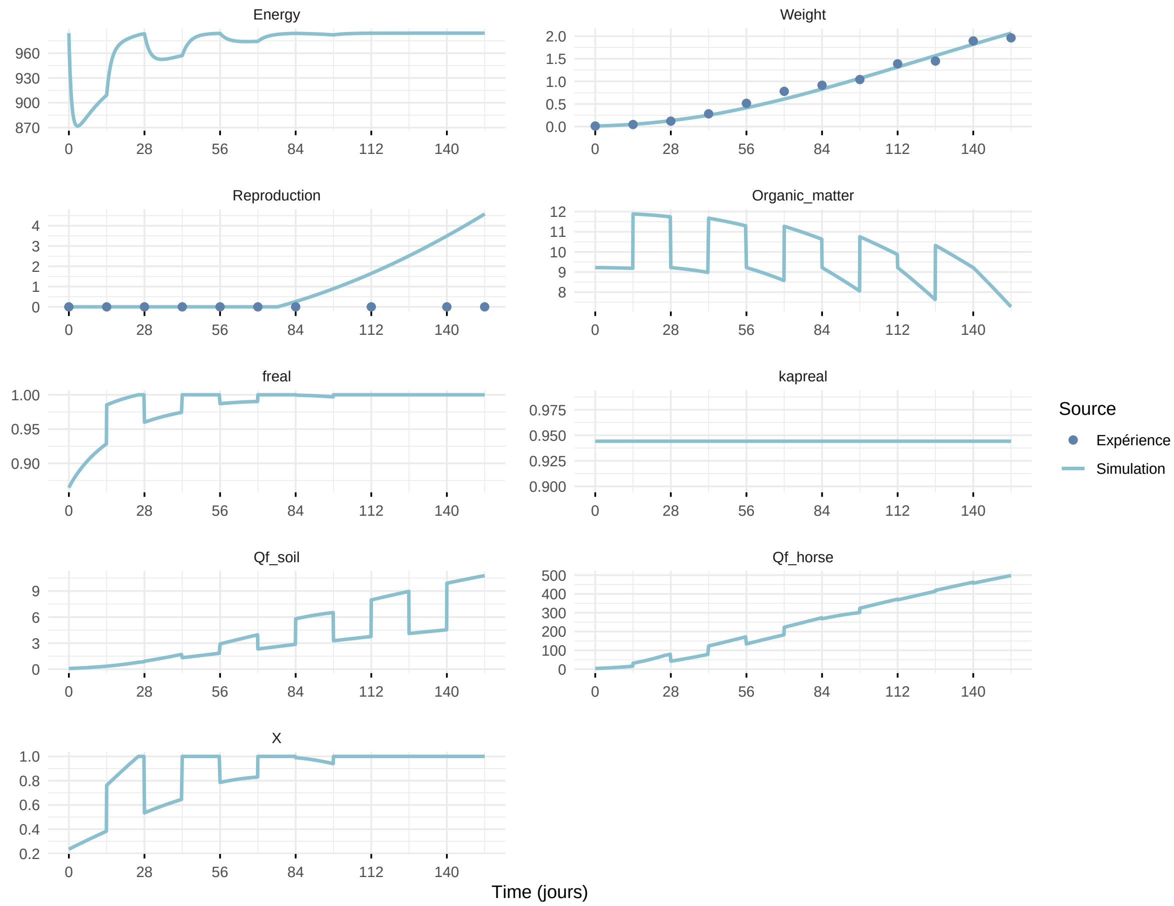
Expérience : Bart2019_XP4_2

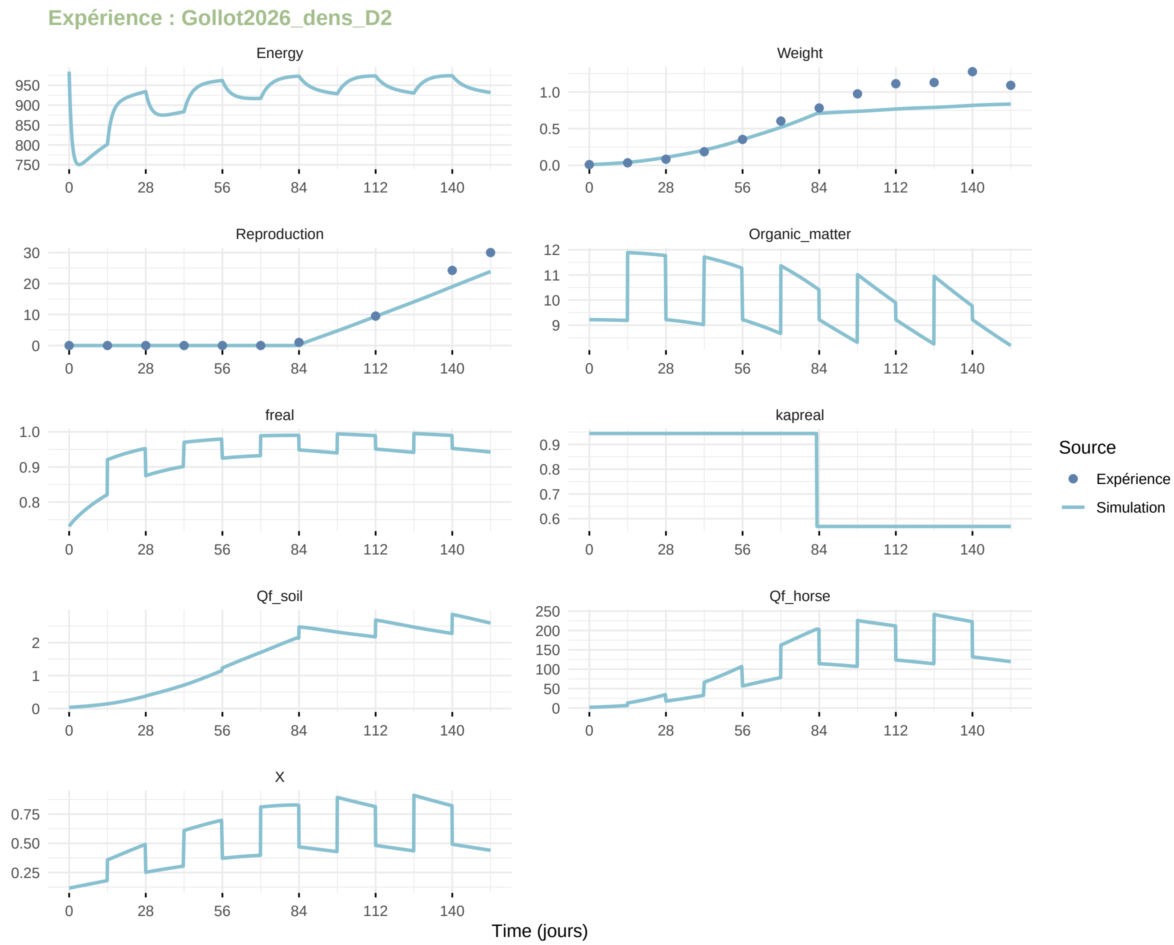


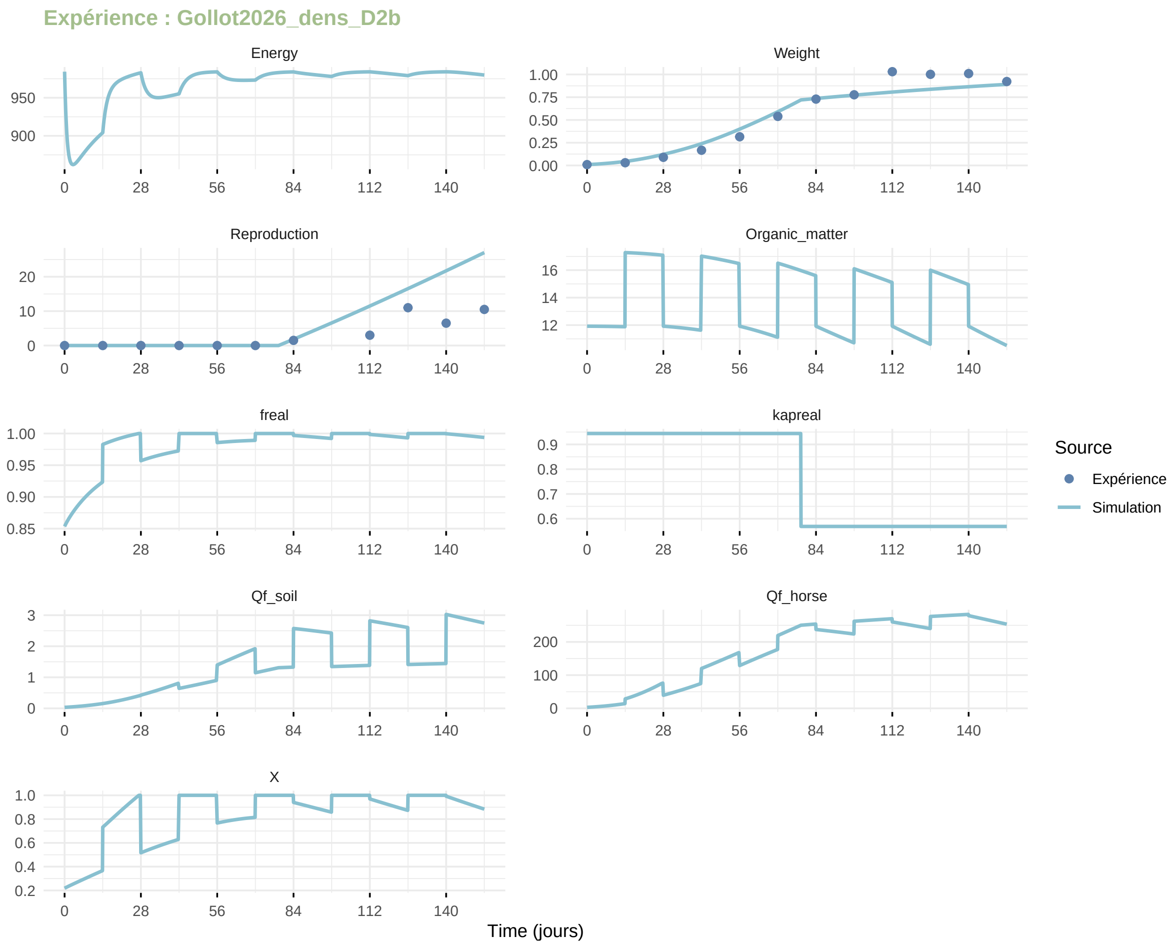

Expérience : Bart2019_XP5
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)


Expérience : Bart2020
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)

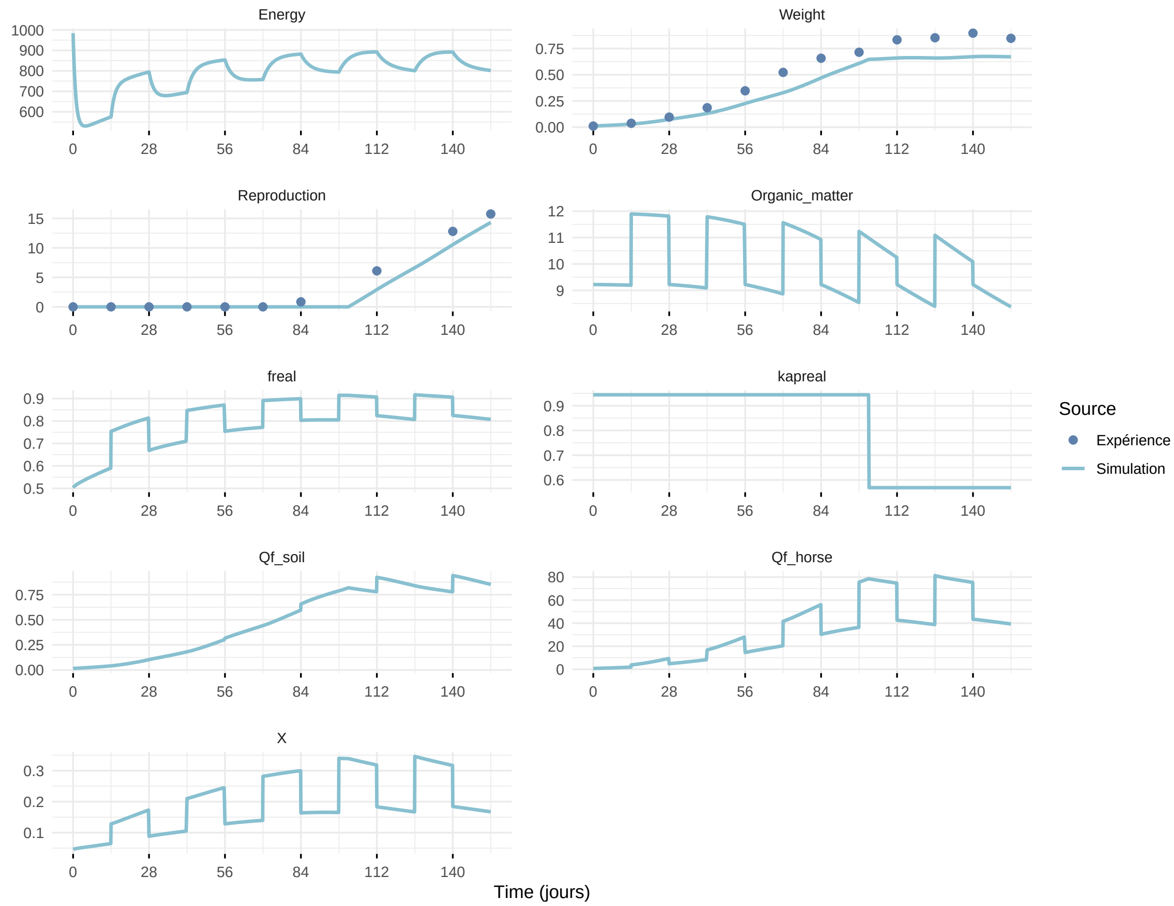
Expérience : Gollot2026_dens_D1




Expérience : Gollot2026_dens_D2
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)


Expérience : Gollot2026_dens_D2b
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Time (jours)
Source
Expérience
Simulation

Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7